



EICS

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MAKE
DRIVING INNOVATION IN MANUFACTURING



Reclaiming Knowledge Structuring from Automation

Human-AI Co-Engineering of KC Models

Seppe Vandenberg, Gilles Eerlings, Gustavo Roveló Ruiz, and Davy Vanacken

Personalized Learning Requires Understanding the Learner

“Two things are crucial for an adaptive system to work: The existence of a means to adapt the task, and the ability to detect the need for adaptation.” ~ Johnson and Taatgen [1]

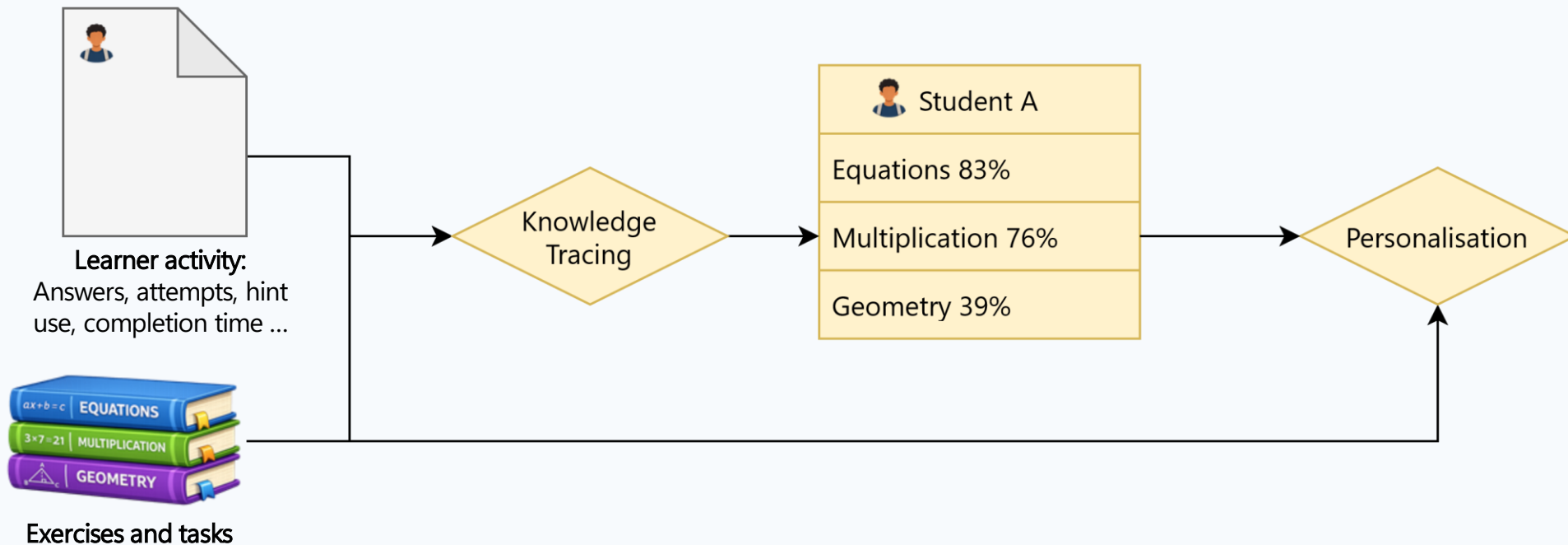


Comprehends	Student A	Student B	Student C
Equations	✓	X	✓
Multiplication	✓	✓	X
Geometry	X	✓	X

Needs extra help with	Geometry	Equations	Multiplication
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Knowledge Tracing (KT) Estimates Learner Knowledge

KT transforms learner interactions into information for personalization



Knowledge Components (KCs) represent skills and concepts a learner can master

Knowledge Components in a Q-Matrix

KC1: Equations

KC2: Multiplications

KC3: Geometry

Knowledge Components in a Q-Matrix

Q1: $3 * 8 = ?$

KC1: Equations	X
KC2: Multiplications	✓
KC3: Geometry	X

Knowledge Components in a Q-Matrix

Q1: $3 * 8 = ?$

Q2: $x + 3 = 5$,
 $x = ?$

KC1: Equations	X	✓
KC2: Multiplications	✓	X
KC3: Geometry	X	X

Knowledge Components in a Q-Matrix

Q1: $3 * 8 = ?$

Q2: $x + 3 = 5$,
 $x = ?$

Q3: Calculate the area
of a square

KC1: Equations

X

✓

X

KC2: Multiplications

✓

X

✓

KC3: Geometry

X

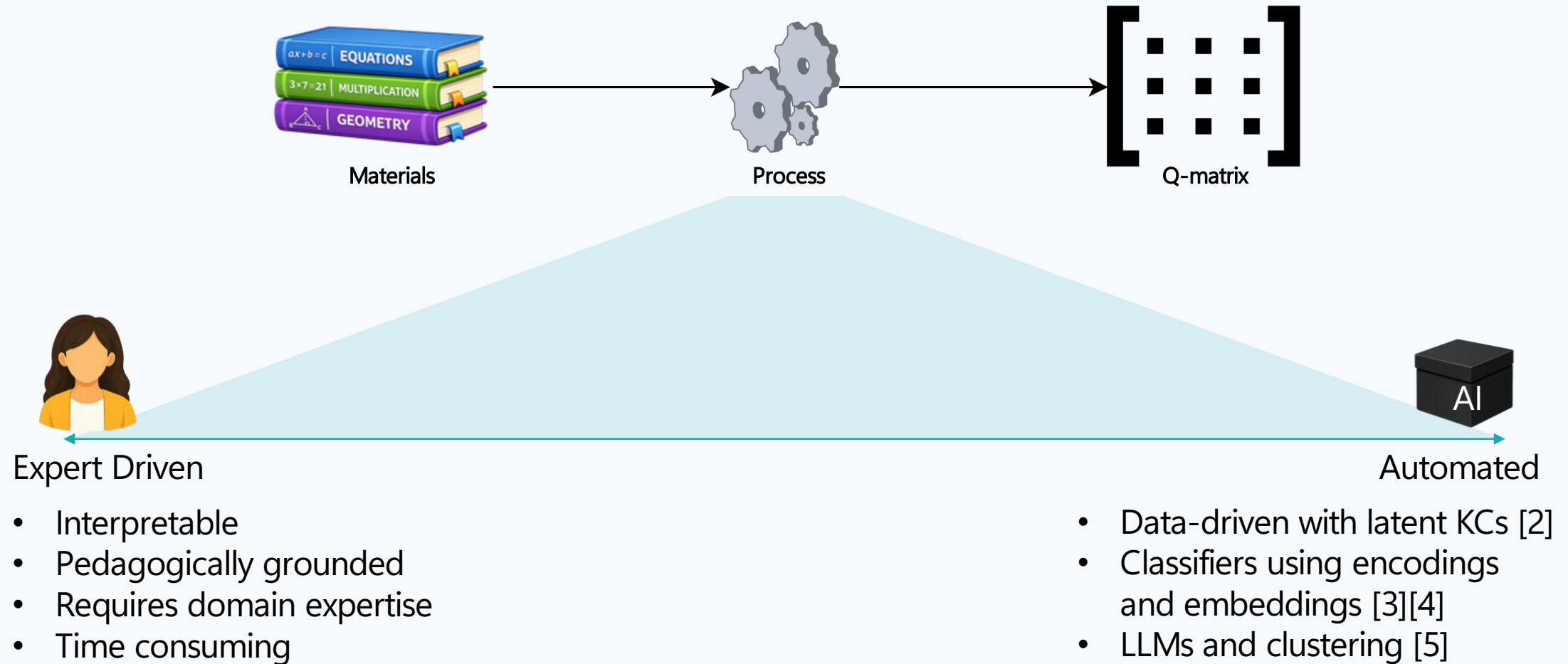
X

✓

Knowledge Components in a Q-Matrix

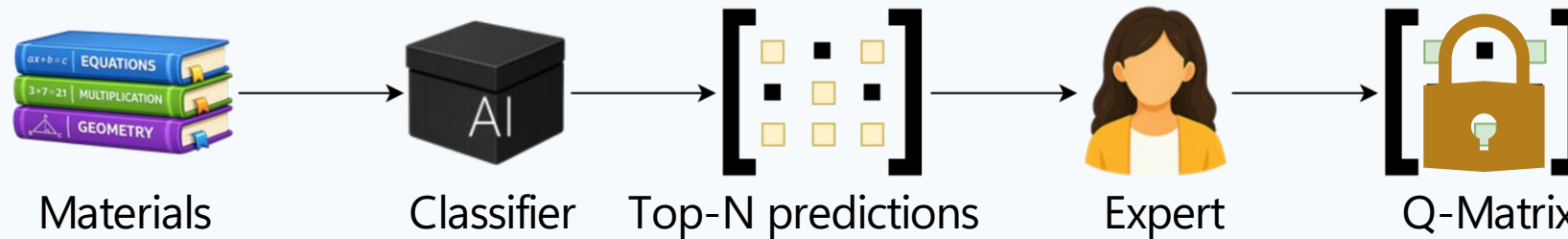
	Q1: $3 * 8 = ?$	Q2: $x + 3 = 5$, $x = ?$	Q3: Calculate the area of a square
KC1: Equations	0	1	0
KC2: Multiplications	1	0	1
KC3: Geometry	0	0	1

How Are Q-Matrices Constructed Today?



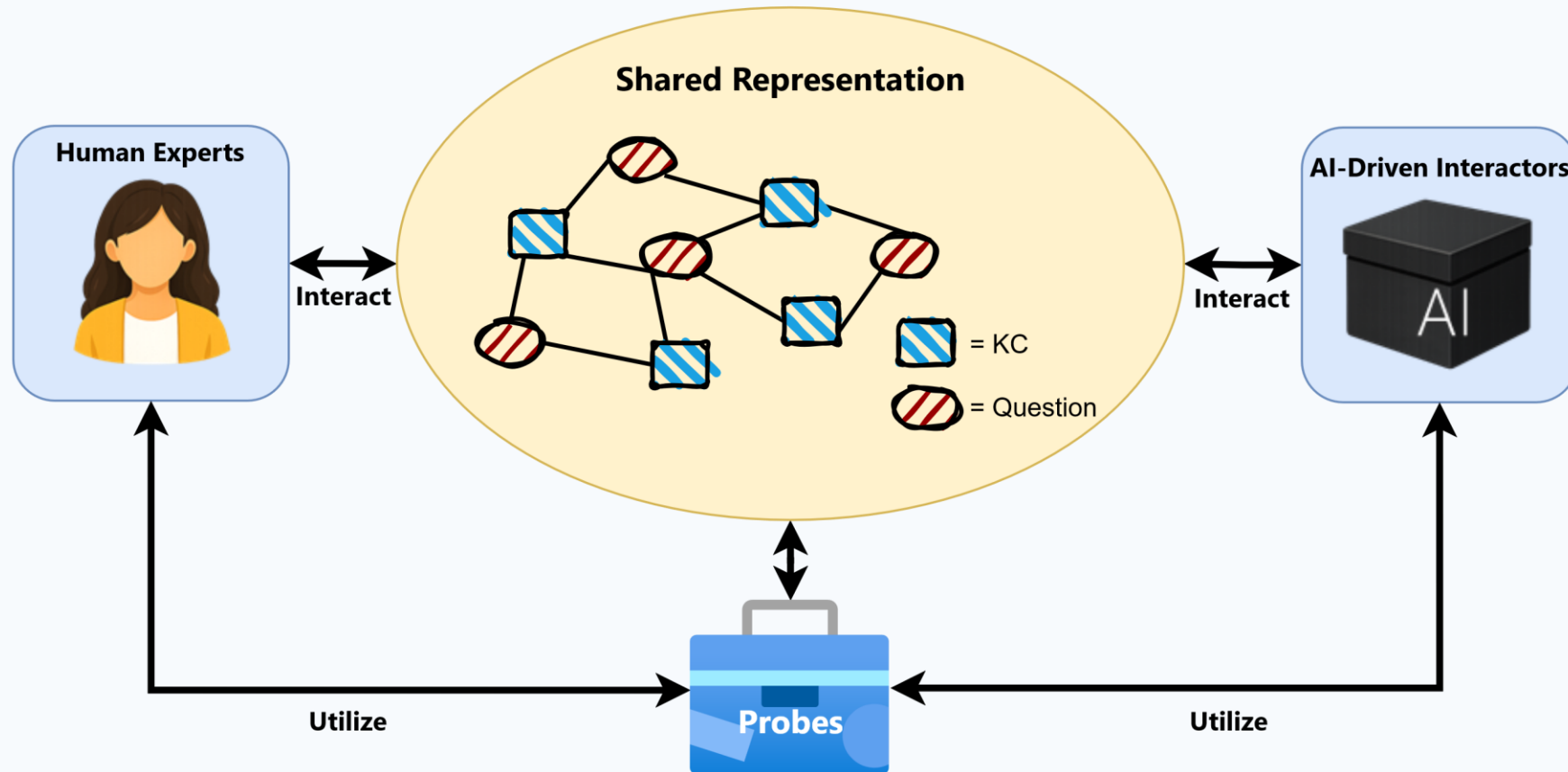
“Human-in-the-Loop” is not Enough

The **only** HITL approach presents experts with AI-generated KC suggestions.



Existing approaches focus on constructing the Q-matrix.
We argue the challenge is engineering a better Q-matrix.

Framework Overview



Continuous refinement instead of one-time construction

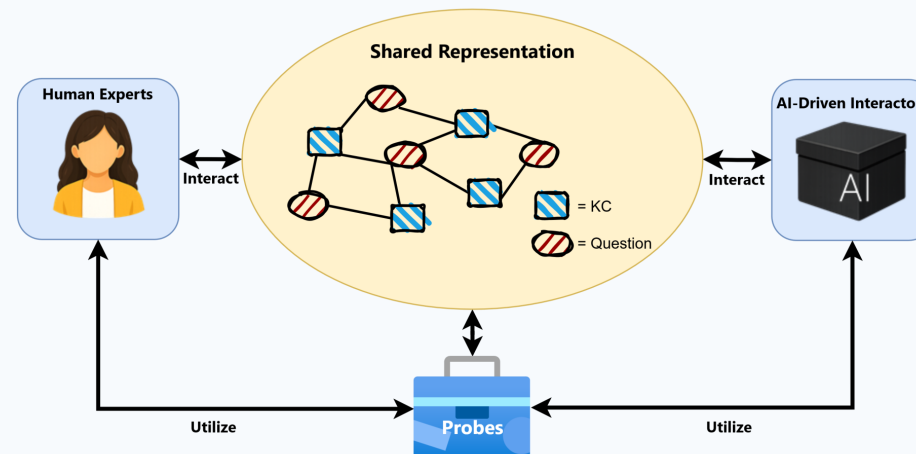
Framework - Interactors

Human Experts

- Use domain expertise
- Refine KCs and mappings
- Resolve automated proposals
- Document rationale

AI-Driven Interactors e.g. LLMs, classifiers, attentive mechanisms, ...

- Generalize across problems
- Spot unusual mappings
- Suggest candidate revisions
- Learn with human feedback



Probes Support Human-AI Collaboration

Explanation Probe

Why was this mapping suggested?

Why should we split this KC?

Simulation Probe

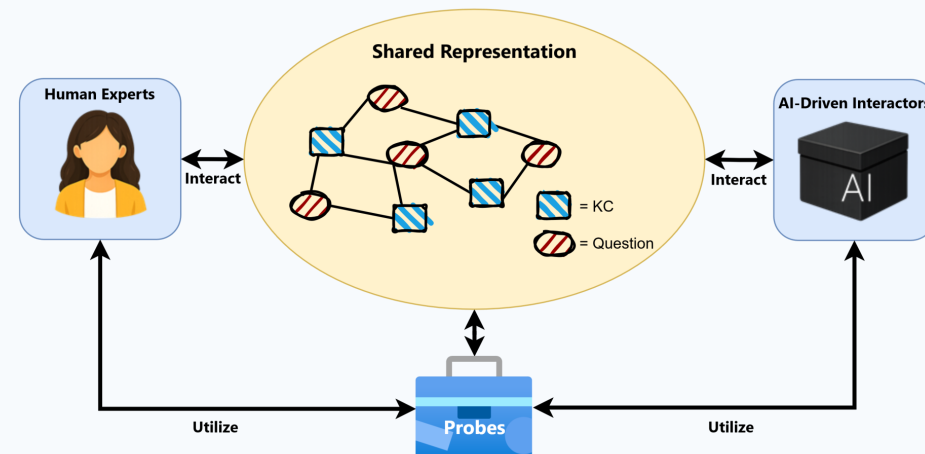
What happens if we add this mapping?

What changes if I merge these KCs?

Inspection Probe

What other inconsistencies still exist?

What are similar problems to this one?



Summary and Next Steps

Key takeaway:

Q-matrix engineering should be treated as a continuous Human-AI collaboration process rather than a one-time construction task.

Next Up:

- Build the prototype
- Evaluate the usefulness of the representation and tools
- Evaluate the resulting Q-matrices with KT benchmarks

Let's Connect!



References

- [1] A. Johnson and N. Taatgen, "User modeling," in *Handbook of human factors in web design*, Mahwah, NJ, US: Lawrence Erlbaum Associates Publishers, 2005, pp. 424–438.
- [2] T. Barnes, "The q-matrix method: mining student response data for knowledge," in *American association for artificial intelligence 2005 educational data mining workshop*, AAAI Press, Pittsburgh, PA, USA, Sep. 2005, p. 8. [Online]. Available: <https://cdn.aaai.org/Workshops/2005/WS-05-02/WS05-02-006.pdf>
- [3] Z. A. Pardos and A. Dadu, "Imputing KCs with Representations of Problem Content and Context," presented at the UMAP '17: 25th Conference on User Modeling, Adaptation and Personalization, Bratislava Slovakia: ACM, Jul. 2017, pp. 148–155. doi: [10.1145/3079628.3079689](https://doi.org/10.1145/3079628.3079689).
- [4] C. S. Tan and J.-J. Kim, "Automated Math Word Problem Knowledge Component Labeling and Recommendation," in *Methodologies and Intelligent Systems for Technology Enhanced Learning, 14th International Conference*, C. Herodotou, S. Papavlasopoulou, C. Santos, M. Milrad, N. Otero, P. Vittorini, R. Gennari, T. Di Mascio, M. Temperini, and F. De la Prieta, Eds., Cham: Springer Nature Switzerland, 2024, pp. 338–348. doi: [10.1007/978-3-031-73538-7_30](https://doi.org/10.1007/978-3-031-73538-7_30).
- [5] H. Moon, R. L. Davis, S. P. Neshaei, and P. Dillenbourg, "Using large multimodal models to extract knowledge components for knowledge tracing from multimedia question information," in *Proceedings of the 18th international conference on educational data mining*, C. Mills, G. Alexandron, D. Taibi, G. L. Bosco, and L. Paquette, Eds., Palermo, Italy: International Educational Data Mining Society, Jul. 2025, pp. 342–353. doi: [10.5281/zenodo.15870171](https://doi.org/10.5281/zenodo.15870171).